Ahmedabad University

CSE400 Fundamentals of Probability in Computing

Project Milestone 4

s1_g7_its

ADAPTIVE TRAFFIC SIGNAL CONTROL USING LAS VEGAS ALGORITHM

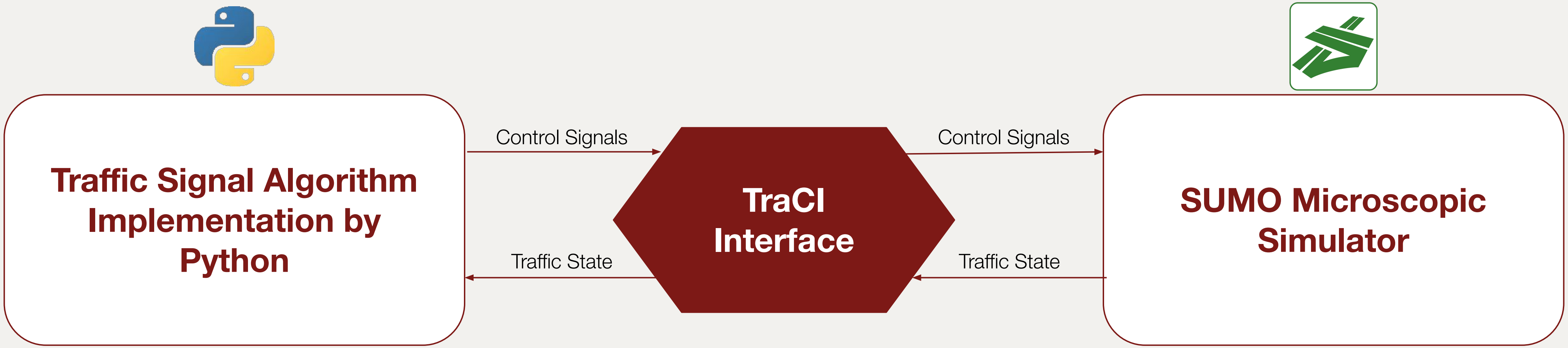
- **Application Domain :**
  - Intelligent Transport System
- **Objective :**
  - Minimizing vehicle waiting time, queue length and congestion at intersection.
- **Baseline :**
  - Deterministic signal timings based on fixed rules.
- **Core Problem :**
  - Signal timing decisions must be made despite uncertain and time-varying traffic conditions.
  - Real-world traffic is stochastic (random arrivals, variable flow)

- **Deterministic Baseline Logic :**
  - Cycle timing determined analytically each cycle.

- **Overview :**
  - Mathematical nature : rule-based (no randomness)
  - Signals operated using :
    - Fixed Thresholds (queue length / waiting time)
    - Static Decision ruling (e.g. max queue gets green)

- **Limitations :**
  - It can not handle sudden traffic booms and uneven arrivals.
  - Some lanes may remain underserved (starvation issue).
  - The system settles for locally optimal decisions, not the best overall solution.

- **Traffic Signal Control Simulation Pipeline :**
  - SUMO simulates microscopic traffic dynamics at the intersection
  - Python Controller implements traffic signal decision logic
  - TraCI Interface enables real-time bidirectional communication



- **Decision Module :**
  - Signal selection is performed at each discrete time-step
  - This is where,
    - Deterministic or randomized algorithm is applied.

- **Sources of uncertainty :**
    - **Random Vehicle Arrivals:** Modeled using Poisson distribution
    - **Variable Service Time:** Depends on vehicle type and driver behavior
    - **Uneven Traffic Flow:** Sudden congestion in specific lanes
    - **Driver Behavior Variability:** Different driving patterns affect flow
    - **Speed Fluctuations:** Non-uniform speeds due to traffic conditions

- Traffic systems are inherently stochastic, making deterministic control insufficient.

- **Random Variables in this system :**
  - These variables model real traffic stochasticity.

Random Variable	Meaning	Distribution	Units
A_i	Vehicle arrivals	Poisson(λ)	vehicles/time-step
S_i	Service rate	Exponential / Fixed	vehicles/time-step
Q_i	Queue length	Derived	vehicles
W_i	Waiting time	Derived	seconds
X_i	Signal selection	Random choice	index

- **Randomized Algorithm Design :**
  - Approach : Hybrid Las Vegas Algorithm
    - Step 1: Identify top candidate lanes (high queue)
    - Step 2: Randomly select among them
    - Step 3: Validate decision (avoid poor choices)
    - Step 4: Fallback → deterministic if needed

- **Difference from baseline :**

Deterministic	Randomized
Always max queue	Probabilistic selection
Predictable	Adaptive
Rigid	Flexible

- **Probabilistic Reasoning :**
  - Traffic arrivals are modeled as a Poisson process, reflecting inherent randomness in real-world flow.
  - Deterministic approaches fail to account for this variability, often leading to suboptimal decisions.

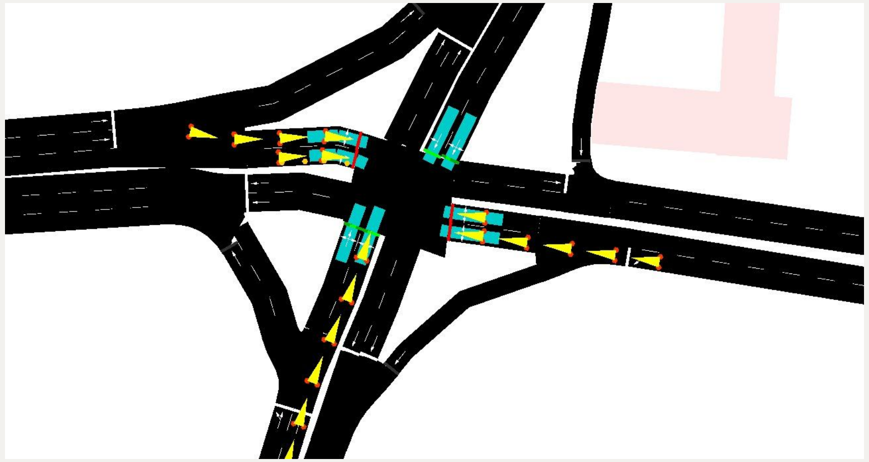
- **Why Randomization is Effective :**
  - A deterministic algorithm always follows the same sequence of steps, and the adversary knows what steps the algorithm takes.
  - By contrast, a randomized algorithm is an algorithm that makes use of a random sequence of bits in deciding what to do next.

• Simulation Implementation :

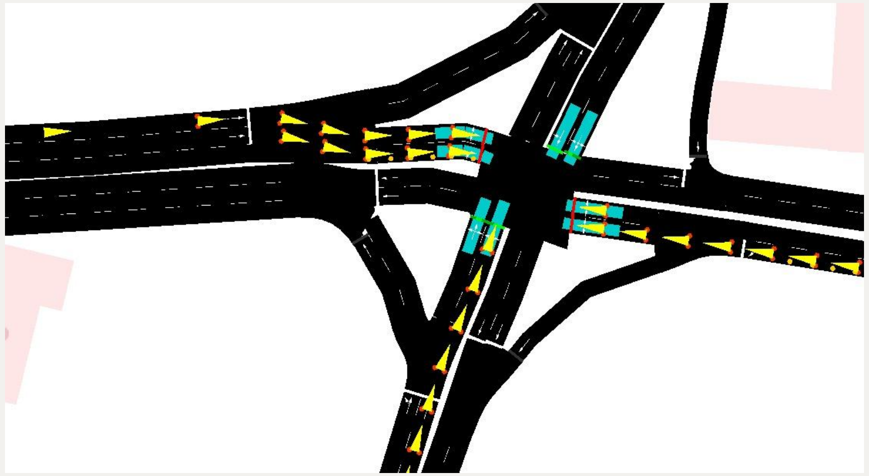
- Code + SUMO

```
def las_vegas_step(Q, trials=30):
    new_Q = []
    total_wait = 0
    best_deltas = []
    for i in range(LANES):
        arrivals = sample_arrivals(lambda_i[i])
        det_departures = min(Q[i], int(mu_i[i]))
        det_Q = max(Q[i] + arrivals - det_departures, 0)
        best_Q = float('inf')
        best_delta = 1
        for _ in range(trials):
            delta = sample_delta_G()
            departures = min(Q[i], int(mu_i[i] * max(delta,1)))
            Qi_candidate = max(Q[i] + arrivals - departures, 0)
            if Qi_candidate < best_Q:
                best_Q = Qi_candidate
                best_delta = delta
        lv_departures = min(Q[i], int(mu_i[i] * max(best_delta,1)))
        lv_Q = max(Q[i] + arrivals - lv_departures, 0)
        Qi = min(det_Q, lv_Q)
        Wi = Qi / max(lambda_i[i], 1e-6)
        total_wait += Wi
        new_Q.append(Qi)
        best_deltas.append(best_delta)
    return new_Q, total_wait / LANES, best_deltas
```

Las vegas algorithm module in python code



Sumo simulation for deterministic cycle

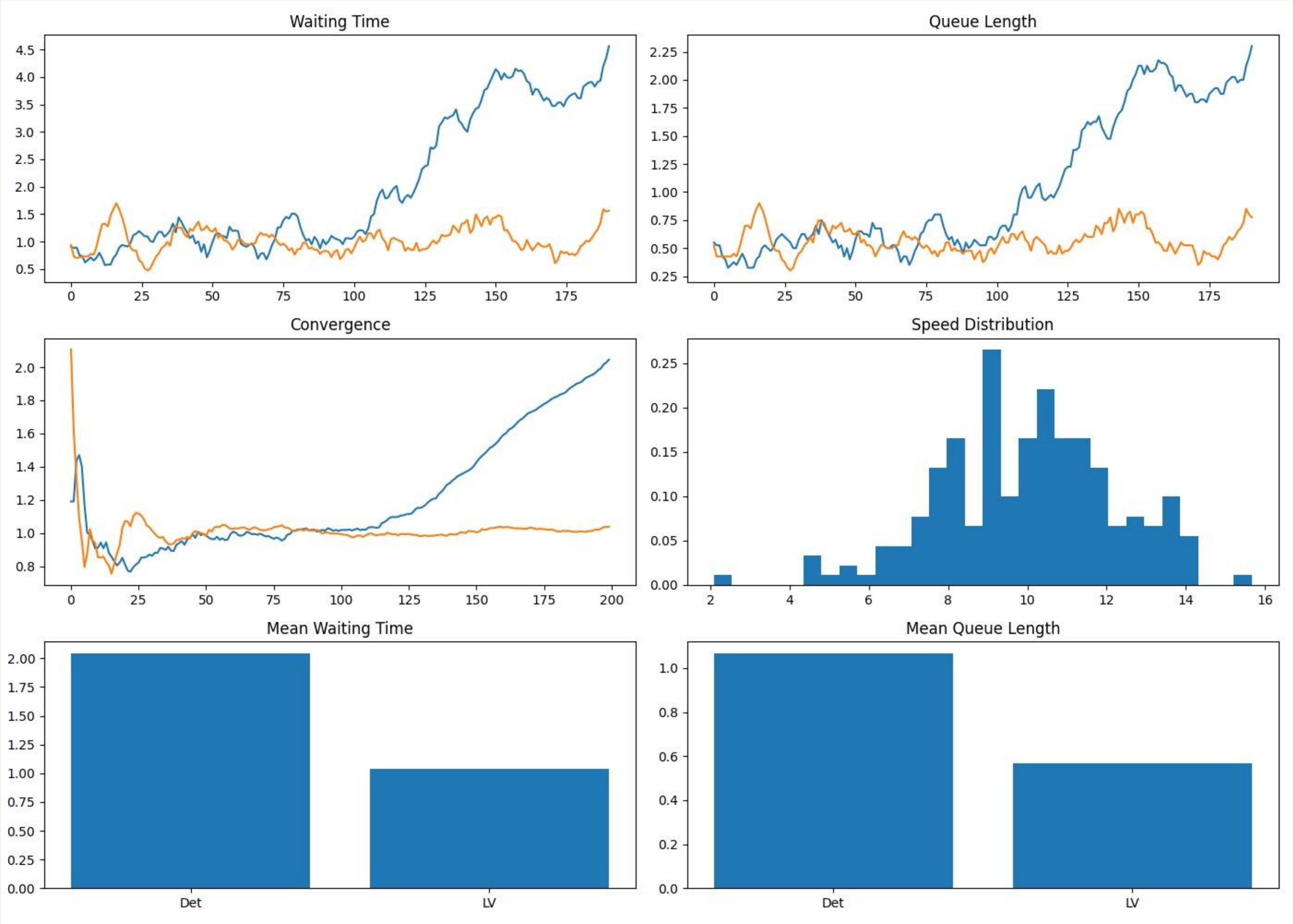


Sumo simulation for Las Vegas cycle

- **Evaluation Metrics :**
  - The metrics used for the evaluation is,

Metric	What it Measures	Why it Matters
Waiting Time (per vehicle)	Time a vehicle spends before crossing	Direct indicator of delay experienced by users
Queue Length	Number of vehicles waiting at a signal	Reflects congestion level at intersections
Convergence Behavior	Stability of system over time	Shows whether the algorithm reaches steady performance
Speed Distribution	Spread of vehicle speeds	Indicates smoothness and flow efficiency

- **Simulation Implementation :**
  - Results




```
===== FINAL RESULT =====
Deterministic : 2.05
Las Vegas     : 1.04
Improvement   : 49.27%
```

Ahmedabad University

- **Key Insights :**
  - Randomization improves adaptability under dynamic and uncertain traffic conditions.
  - Helps prevent lane starvation by avoiding strict priority-based decisions.
  - Achieves lower waiting time and queue length compared to deterministic approach.
  - Hybrid (controlled randomization) provides better stability than pure Las Vegas.
  - System shows faster convergence and consistent performance over time.

- **Challenges :**
  - **Stability vs Randomness:** Excess randomness causes instability; too little reduces effectiveness
  - **Real-Time Constraints:** Decisions must be computed quickly with minimal overhead
  - **Reproducibility:** Results vary across runs; requires multiple simulations and averaging
  - **Scalability:** Extending to multi-intersection systems increases complexity
  - **System Integration:** TraCI communication delays and synchronization issues

Thank You

Rushin Modi - AU2440028

Viranshul Panchal - AU2440209

Harchol-Balter, M. (2024). Introduction to probability for computing. Cambridge University Press. March 25, 2026, https://www.cs.cmu.edu/~harchol/Probability/chapters/chpt21.pdf

M. E. M. Ali, A. Durdu, S. A. Celtek and A. Yilmaz, "An Adaptive Method for Traffic Signal Control Based on Fuzzy Logic With Webster and Modified Webster Formula Using SUMO Traffic Simulator," in IEEE Access, vol. 9, pp. 102985-102997, 2021, doi: 10.1109/ACCESS.2021.3094270.